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Assignment 2 Q-Learning on Taxi-v3: Hyperparameter Study and Analysis

1. Introduction

This report presents an experimental study of tabular Q-learning applied to the Taxi-v3 environment. The goal is to analyze how the learning rate (α) and exploration factor (ε) affect learning performance, convergence speed, and stability. Performance is evaluated using two metrics: return per episode and steps per episode. Based on the experimental results, the best-performing parameter configuration is selected and re-run to validate the choice.

2. Environment and Algorithm

The Taxi-v3 environment is a discrete grid-based problem in which an agent must pick up a passenger at one location and drop them off at a target destination. The state space is finite and discrete, making it suitable for tabular Q-learning. The agent follows an ε-greedy policy for action selection.

Q-learning updates the action-value function according to:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \, [\, r + \gamma \max_{a'} Q(s',a') - Q(s,a) \,]$$

where α is the learning rate, γ is the discount factor (fixed at 0.9 in all experiments), and ε controls the exploration rate.

3. Methodology

All experiments were conducted on the Taxi-v3 environment using tabular Q-learning. Each configuration was trained for **5,000 episodes** with a maximum of **200 steps per episode**. A fixed random seed was used for reproducibility. For each parameter configuration, one full training run was performed.

The following metrics were recorded during training:

- Return per episode
- Steps per episode

The following parameter configurations were evaluated:

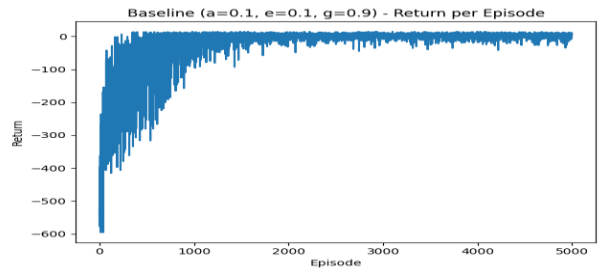
- **Baseline:** α = 0.1, ε = 0.1, γ = 0.9

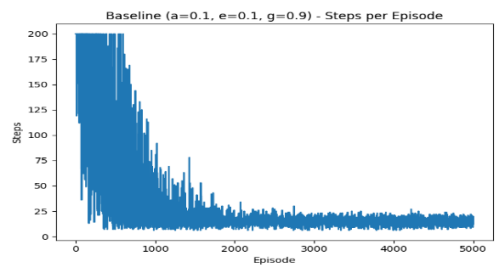
- **Learning rate sweep:** $\alpha \in \{0.001, 0.01, 0.2\}$ with $\varepsilon = 0.1$, $\gamma = 0.9$
- **Exploration sweep:** $\varepsilon \in \{0.2, 0.3\}$ with $\alpha = 0.1$, $\gamma = 0.9$

After training, a greedy evaluation phase was performed and summary statistics were recorded.

4. Results and Reflections

4.1 Baseline ($\alpha = 0.1$, $\varepsilon = 0.1$, $\gamma = 0.9$)

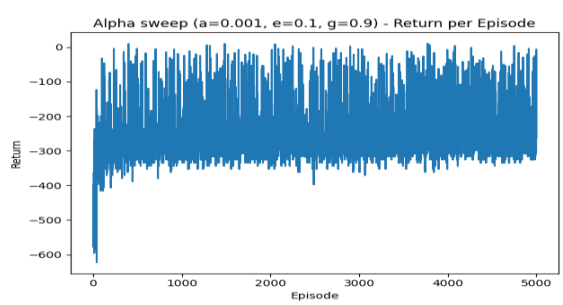


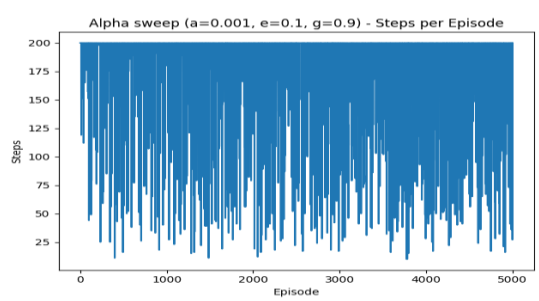


Reflection:
The return curve starts highly negative and noisy, indicating that the agent initially behaves almost randomly and accumulates many penalties during exploration. Over time, the return increases and stabilizes, showing that the agent learns a more effective policy. At the same time, the steps-per-episode curve decreases significantly, meaning the agent completes the task using fewer actions. The simultaneous improvement in return and reduction in steps demonstrates that the agent is not only achieving higher rewards but also doing so more efficiently. The reduced variance in both curves toward later episodes indicates convergence to a stable policy.

4.2 Learning Rate Sweep

$\alpha = 0.001$ ($\varepsilon = 0.1$, $\gamma = 0.9$)

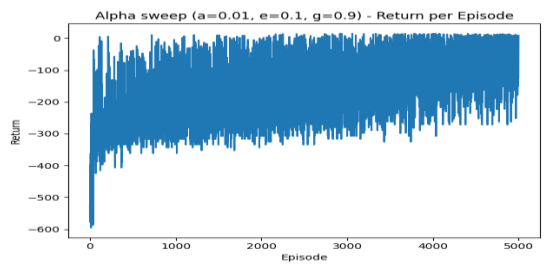


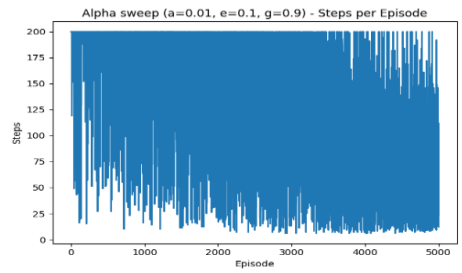


Reflection:
With a very small learning rate, the return remains highly noisy and improves only slowly over time, indicating that the agent updates its Q-values very cautiously. Although there is a

slight upward trend, convergence is extremely slow and unstable. The steps-per-episode curve remains high for most of training, showing that the agent continues to take inefficient and lengthy paths. This demonstrates that too small a learning rate prevents the agent from adapting quickly enough to the environment, resulting in poor learning speed and weak performance.

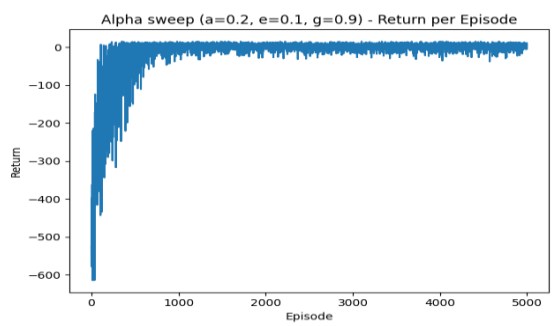
α = 0.01 (ε = 0.1, γ = 0.9)

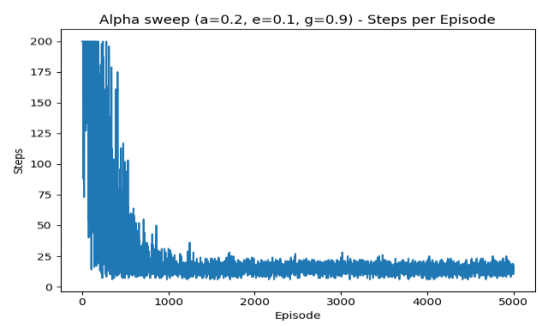




Reflection:
Compared to α = 0.001, the return increases more steadily and reaches a higher level earlier in training, showing improved learning speed. The steps per episode gradually decrease, indicating that the agent is learning more efficient routes. However, both curves still exhibit noticeable variance for a long portion of training, suggesting slower and less stable convergence. This setting improves learning compared to a very small α, but remains suboptimal in terms of convergence speed and stability.

α = 0.2 (ε = 0.1, γ = 0.9)

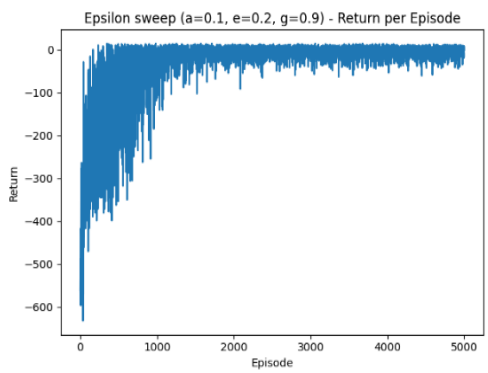


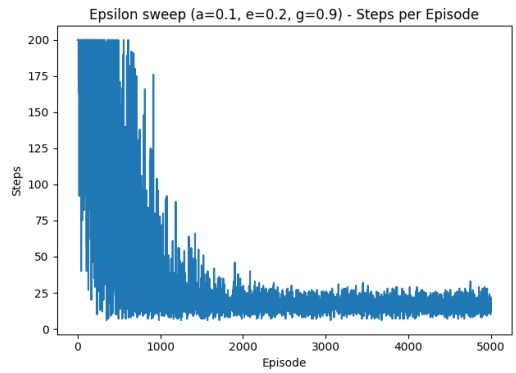


Reflection:
With a higher learning rate, the return increases rapidly and stabilizes much earlier in training, indicating fast learning and strong policy improvement. The steps-per-episode curve drops sharply and converges to a low value, showing that the agent quickly learns efficient paths. The reduced variance in both curves after early training suggests stable convergence. This demonstrates that a moderately large learning rate significantly improves both learning speed and final performance.

4.3 Exploration Rate (ε) Sweep

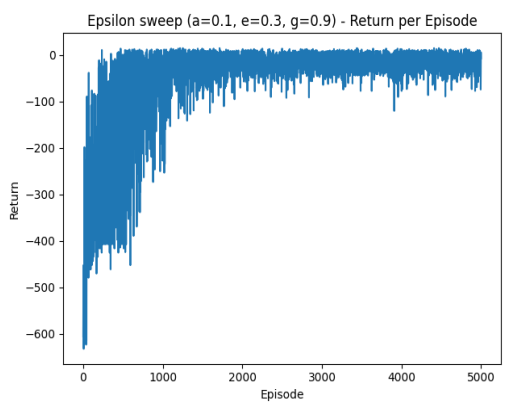
ε = 0.2 (α = 0.1, γ = 0.9)

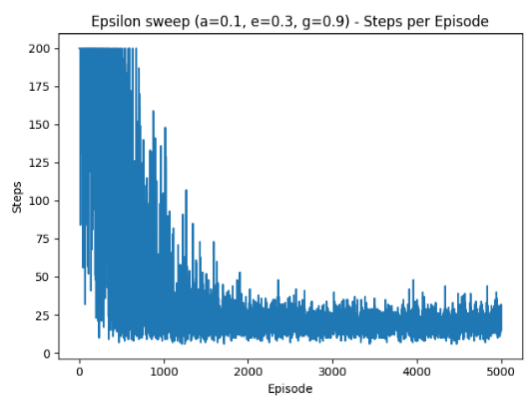




Reflection:
With a higher exploration rate, the return initially improves more slowly and shows higher variance, reflecting increased random exploration. Over time, the return stabilizes at a good level, indicating that sufficient exploration helps the agent discover effective strategies. The steps-per-episode curve decreases steadily but remains more variable compared to the baseline, showing a trade-off between exploration and efficiency. This setting improves robustness but slightly slows convergence.

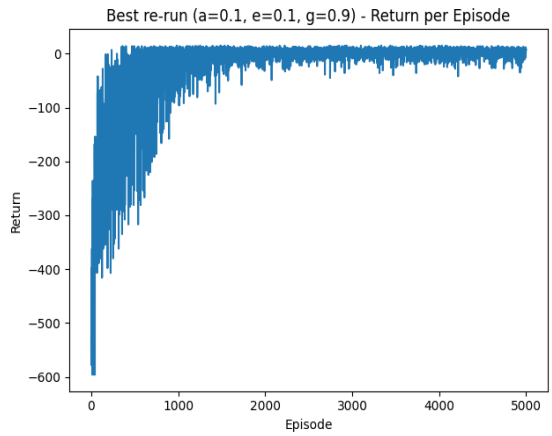
ε = 0.3 (α = 0.1, γ = 0.9)

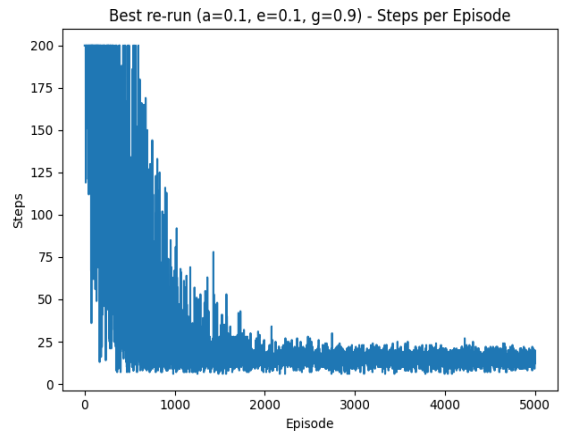




Reflection:
At an even higher exploration rate, the return remains noisier for a longer duration, indicating that excessive exploration delays exploitation of the learned policy. Although the agent eventually achieves reasonable performance, convergence is slower and less stable. The steps-per-episode curve also shows higher variance, confirming that the agent continues to take inefficient paths due to frequent random actions. This demonstrates that too much exploration can hinder learning efficiency.

4.4 Best Parameter Re-Run (α = 0.1, ε = 0.1, γ = 0.9)





Reflection:
Using the best-performing parameter combination, the return improves quickly and stabilizes at a high level, indicating strong and consistent policy performance. The steps-per-episode curve decreases rapidly and remains low, showing that the agent reliably finds short and efficient paths. Compared to other configurations, this run shows faster convergence, lower variance, and better overall stability. This confirms that the chosen balance between learning rate and exploration achieves both efficient learning and strong final performance.

6. Limitations and Future Work

A limitation of this study is that only a single random seed was used per configuration. Future work should average results over multiple independent runs to provide statistically stronger conclusions. Additionally, extending this approach to larger state spaces using function approximation would allow scaling beyond tabular Q-learning.

7. Conclusion

This study demonstrates that both the learning rate and exploration factor significantly influence Q-learning performance. Very small learning rates lead to slow convergence, while moderate learning rates greatly improve learning speed. Higher exploration increases variance and slows convergence. The selected configuration (α = 0.1, ε = 0.1) provides the best balance between speed, stability, and final performance.

8. Five Talking Points

1. I implemented and evaluated tabular Q-learning on the Taxi-v3 environment using systematic sweeps of learning rate and exploration rate.
2. I measured both return per episode and steps per episode to evaluate learning quality and efficiency.
3. The experiments demonstrate the trade-off between fast learning, stability, and exploration-induced variance.
4. The learning curves show clear convergence behavior, with decreasing variance over time.
5. Based on quantitative metrics and convergence behavior, I selected and validated the best-performing parameter configuration.